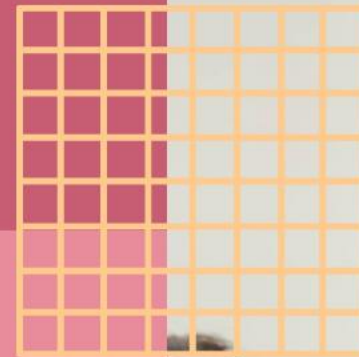


HEALTH AND GENES

Reading skill: Scanning for details





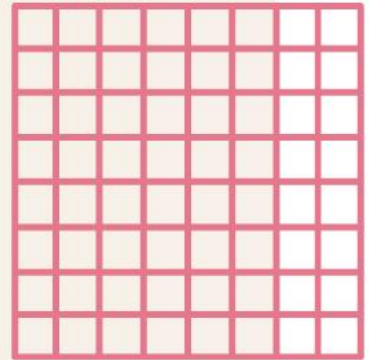
AGENDA

1.Scanning for details and Concept Maps

2.Discussion

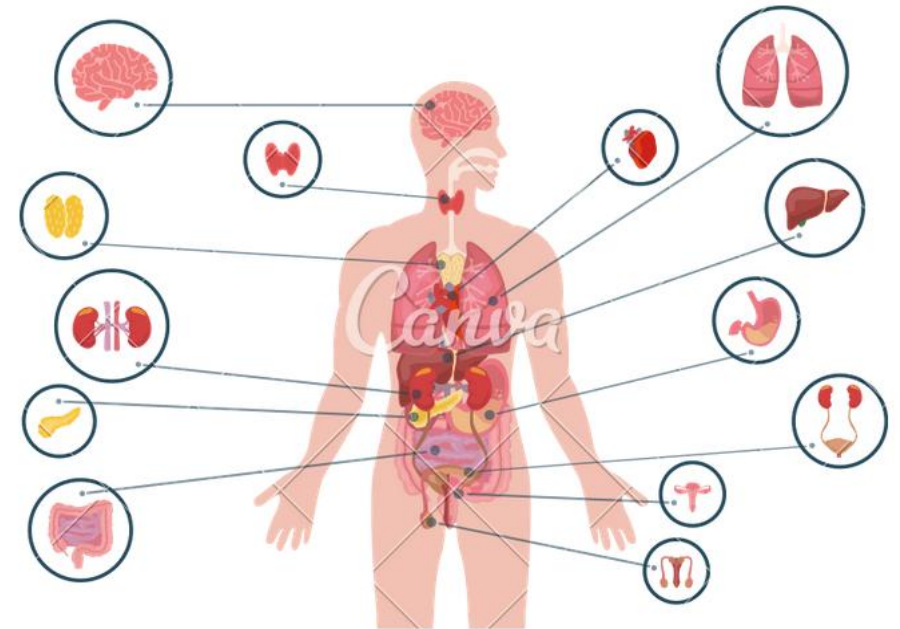
3.Vocabulary building

4.Reading



DO YOU KNOW YOUR BODY WELL? Warm Up

LET'S CHECK IT WITH A QUIZ!

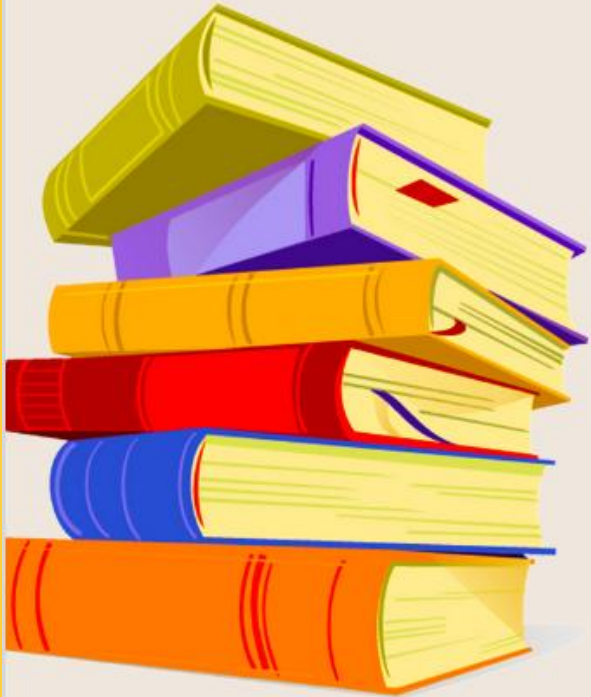




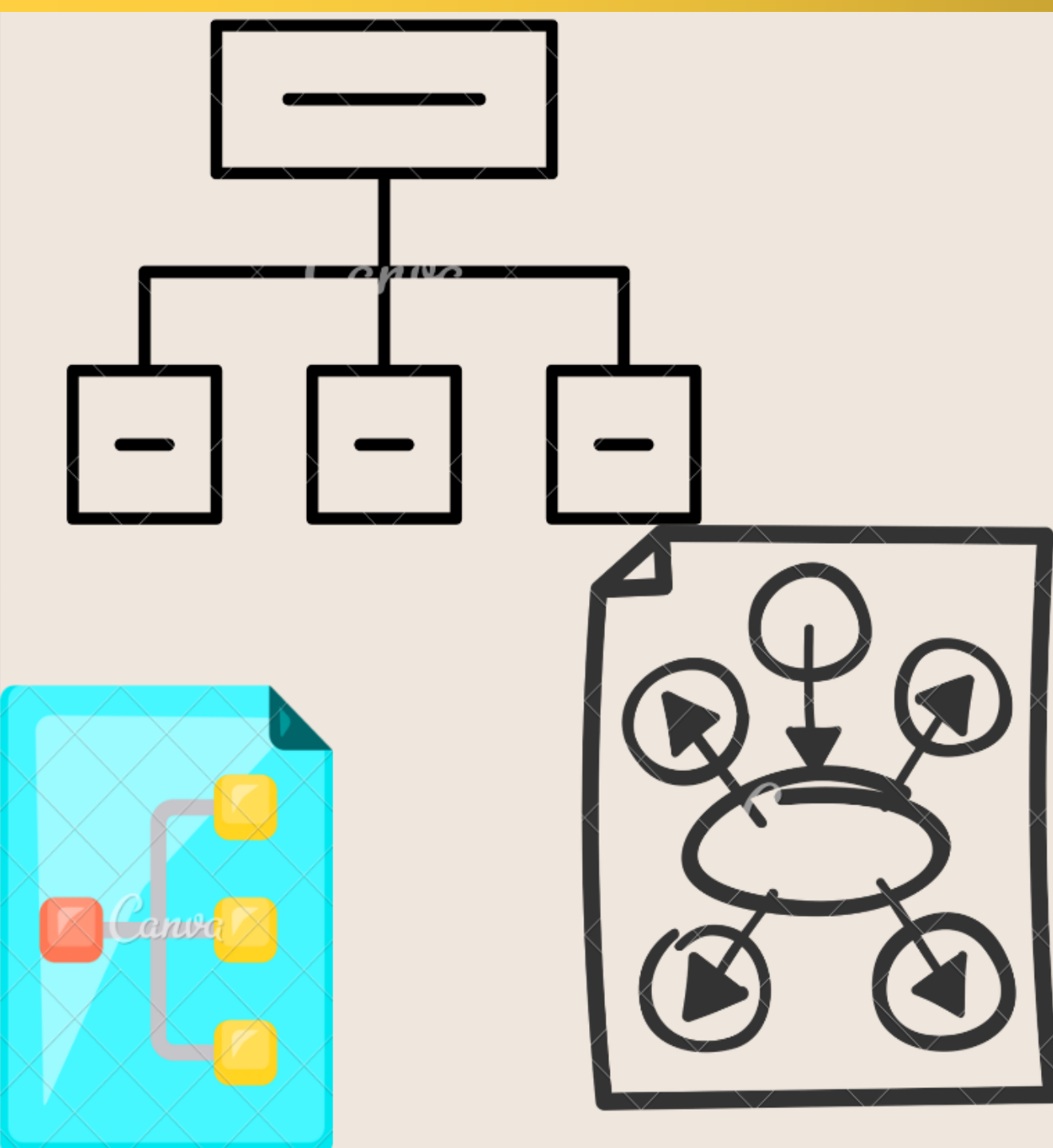
Reading Strategies



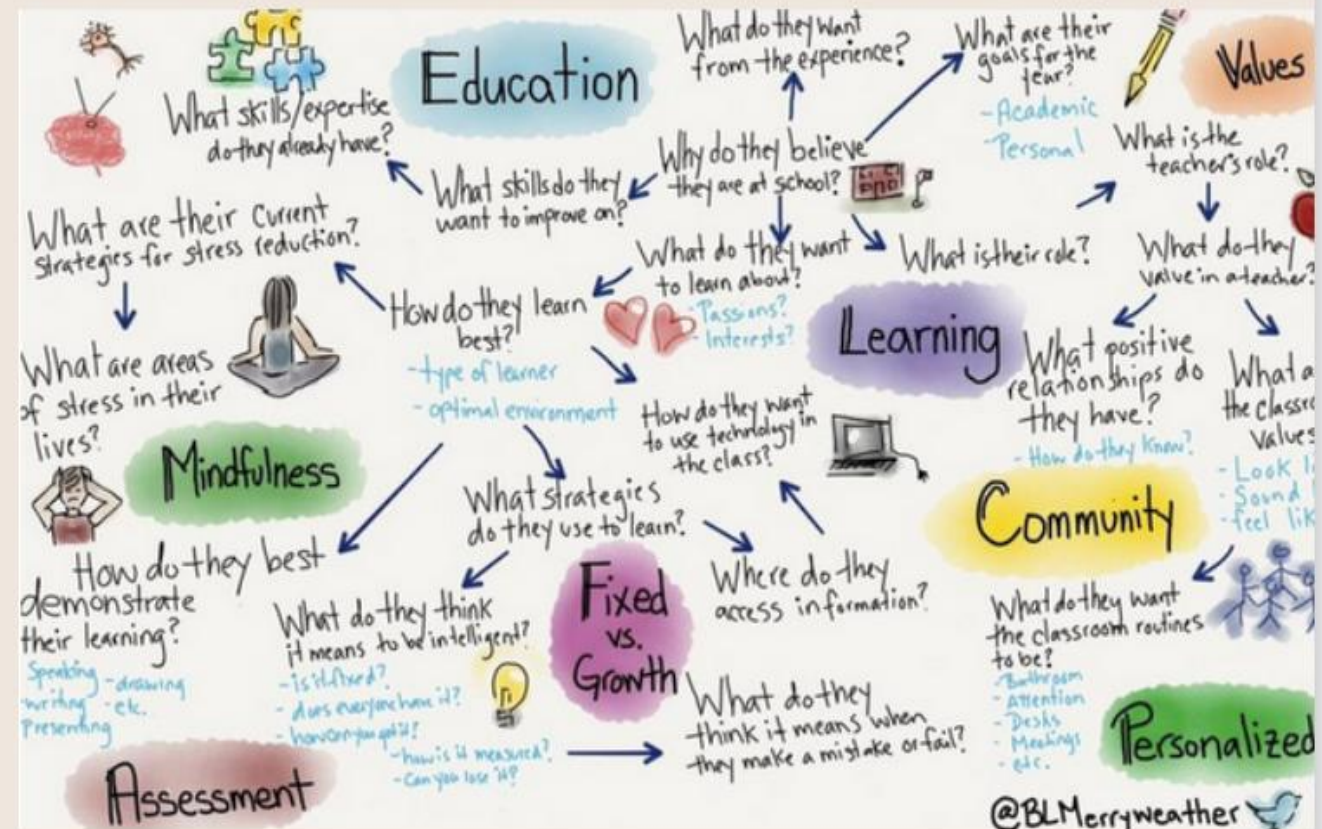
WHAT IS "SCANNING" ?



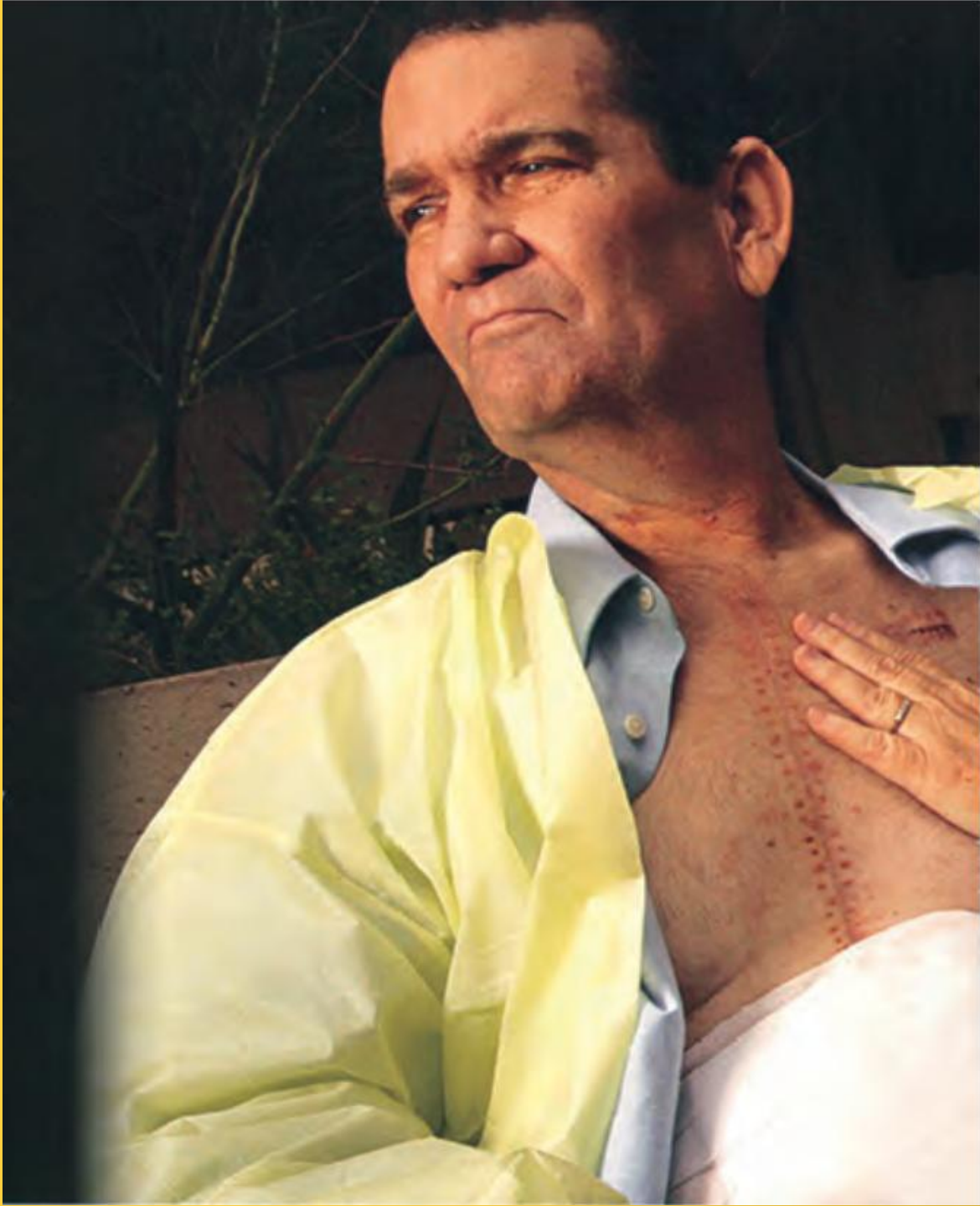
let's find out more by
watching a video



Concept maps are visual representations of information. They can take the form of **charts, graphic organizers, tables, flowcharts, Venn Diagrams, timelines, or T-charts.** . They are a powerful study strategy because they help you **see the big picture:** by starting with higher-level concepts, concept maps help you chunk information based on **meaningful connections.** In other words, knowing the big picture makes details more significant and easier to remember.



@BLMerryweather



MENDING BROKEN
HEARTS

In pairs, discuss the following and share your ideas.

- . What are the most common illnesses affecting people in your country? Why do you think they are common?
- . Do you know anyone who suffers from allergies? What do you think causes them?
- . Do you think a person's health is largely due to his or her lifestyle and environment, or is it decided from the time he or she is born?



- 

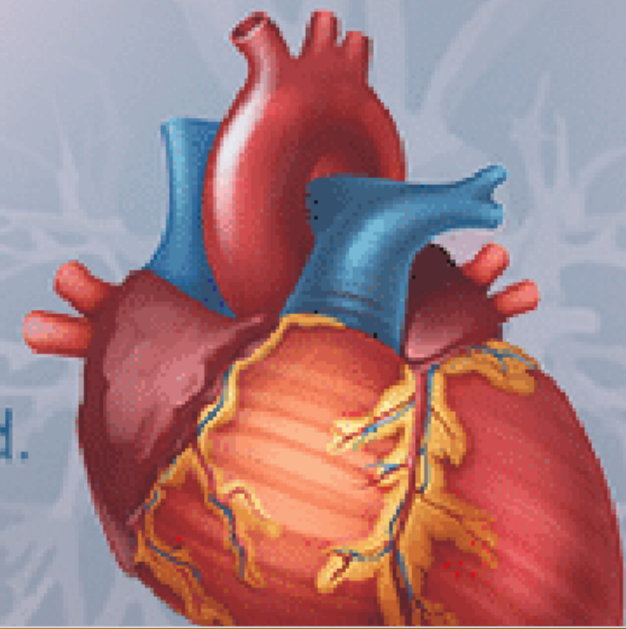




**Who do you think these
people are?**

Interesting facts about the human heart

- ✓ The heart is just the size of your closed fist, but as enough power to pump nearly 7571 liters of blood everyday
- ✓ Your heart beats 100000 times a day without rest
- ✓ A woman's heart beats faster than a man's heart
- ✓ In an average lifetime, Heart pumps approximately 1 million barrels of blood.



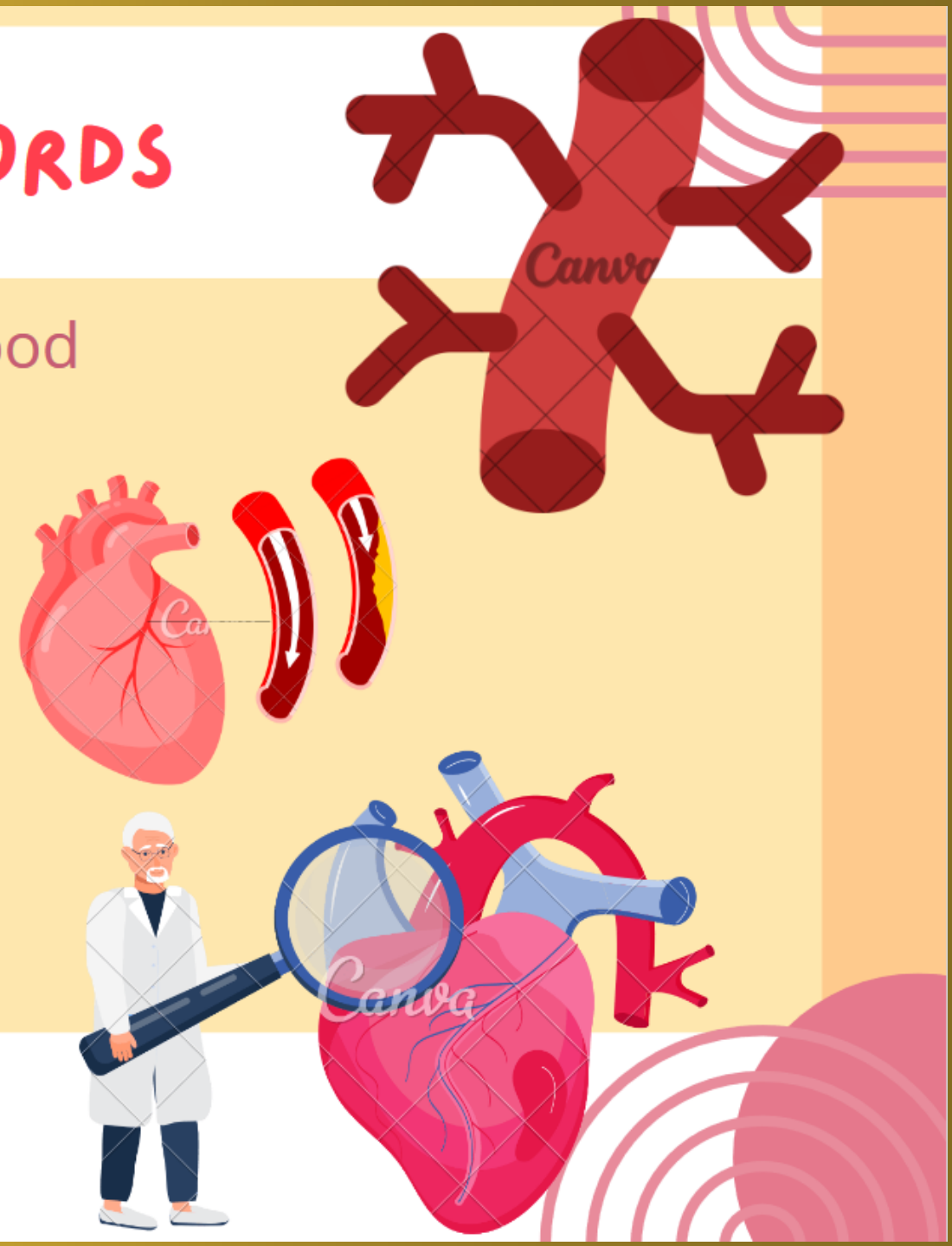
LET'S LEARN NEW WORDS

Arteries - tubes in the body that carry blood to different parts of the body

Cardiovascular- relating to the heart and blood vessels

Coronary- belonging or relating to heart

Cardiologist- doctor who specialises in dealing with heart and its diseases

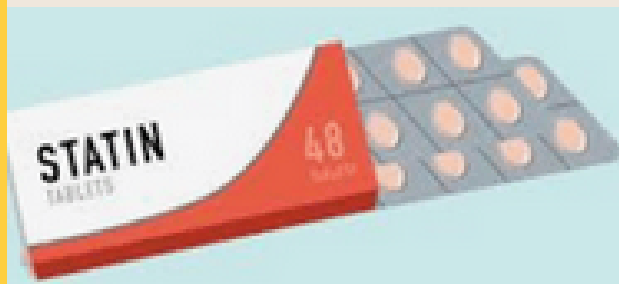




Genetic
Predisposition

Predisposition- If someone has a predisposition to a disease or condition, they are more likely to get that disease or suffer from it.

Foolproof- If smth is foolproof, it is incapable of going wrong or falling to succeed.



Statins- class of drugs used to lower cholesterol level

To compound-
make a problem worse by adding to it

cadaver

noun. a dead body, especially one intended for dissection.



casualty

noun. someone who is hurt or killed in an accident or war.



atherosclerosis

noun. a disease in which your arteries become hard, which makes it difficult for the blood to flow through.

mutation

noun. a change in the genetic structure of an animal or plant that makes it different from others of the same kind.



Almost there!



Nitroglycerin tablets

noun. can be used as heart medication.



exacerbate

make (a problem, bad situation, or negative feeling) worse

impede

delay or prevent (someone or something) by obstructing them; hinder.



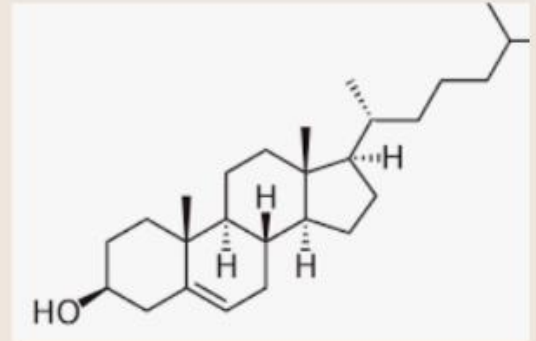
genome

noun. all the genes in one cell of living thing.

You made it !

cholesterol

noun. a chemical substance found in your blood. Too much cholesterol in your body may cause heart disease.



Dysfunctional

adjective. something (in medical situations, an organ or a body part) is not working or functioning normally.



decoy

noun. a decoy is a person or thing used to lure someone into danger. In hunting, it is an image of a bird or an animal used to lure game.



A. **Completion.** Complete the information by circling the correct word in each pair.

Gloria Stevens is lying on her back, staring at an image of her own beating heart. At this moment, she is 1. (**exacerbating** / **submitting**) herself to a life-saving procedure. Her doctor is inserting a thin tube that will pass through a series of arteries to Gloria's heart. It eventually arrives at a spot where plaque has narrowed the arterial channel by 90 percent, severely 2. (**compounding** / **impeding**) the blood passing through it. As Gloria watches on the monitor, the narrowing in her artery disappears, and suddenly a large 3. (**stress** / **volume**) of blood starts flowing, like a river in a flood.

The procedure is over. It has lasted only half an hour. In all likelihood, Gloria will be able to go home the next day. So will a few thousand other patients in the United States undergoing such routine procedures today—more than a million of them a year with a disease that has reached epidemic 4. (**mutations** / **proportions**). Genetic factors may account for some patients' heart disease, but in most cases it is 5. (**compounded** / **submitted**) by a high-fat diet and low levels of activity. Additionally, the 6. (**proportion** / **stress**) of busy modern lifestyles is likely to 7. (**exacerbate** / **impede**) cardiac symptoms.

The procedure has not fully cured Gloria of her coronary atherosclerosis, and she is still 8. (**principal** / **susceptible**) to future blockages and coronary heart disease. But her quality of life will likely improve as a result of treatment. She'll breathe more easily, and maybe she'll live longer, too.



Definitions. Match the correct form of words in **red** from above with their correct definition.

1. to make movement or development difficult _____
2. a feeling of worry and tenseness _____
3. to develop different characteristics _____
4. amount of space something contains _____
5. a partial number compared to a total number _____
6. likely to be affected by something _____
7. first in order of importance _____
8. to allow something to be done _____



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Definitions. Match the correct form of words in **red** from above with their correct definition.

1. to make movement or development difficult _____ **Impede**
2. a feeling of worry and tenseness _____ **stress**
3. to develop different characteristics _____ **mutate**
4. amount of space something contains _____ **volume**
5. a partial number compared to a total number _____ **proportion**
6. likely to be affected by something _____ **susceptible**
7. first in order of importance _____ **principle**
8. to allow something to be done _____ **submit**

Clogged Pipes and Ruptured Walls

Worldwide, coronary heart disease kills 17 million people every year. **Exacerbated** by the export of Western lifestyle—motorized transportation, abundant meat and cheese, workdays conducted from the comfort of a well-padded chair—incidence of the disease is soaring.

Until recently, cardiologists approached heart disease as a plumbing problem. Just as mineral deposits restrict the flow of water through a pipe, a buildup of plaque **impedes** the flow of blood through an arterial channel. Doctors now dismiss this “clogged-pipes model” as an idea whose time has passed. It’s just not that simple.

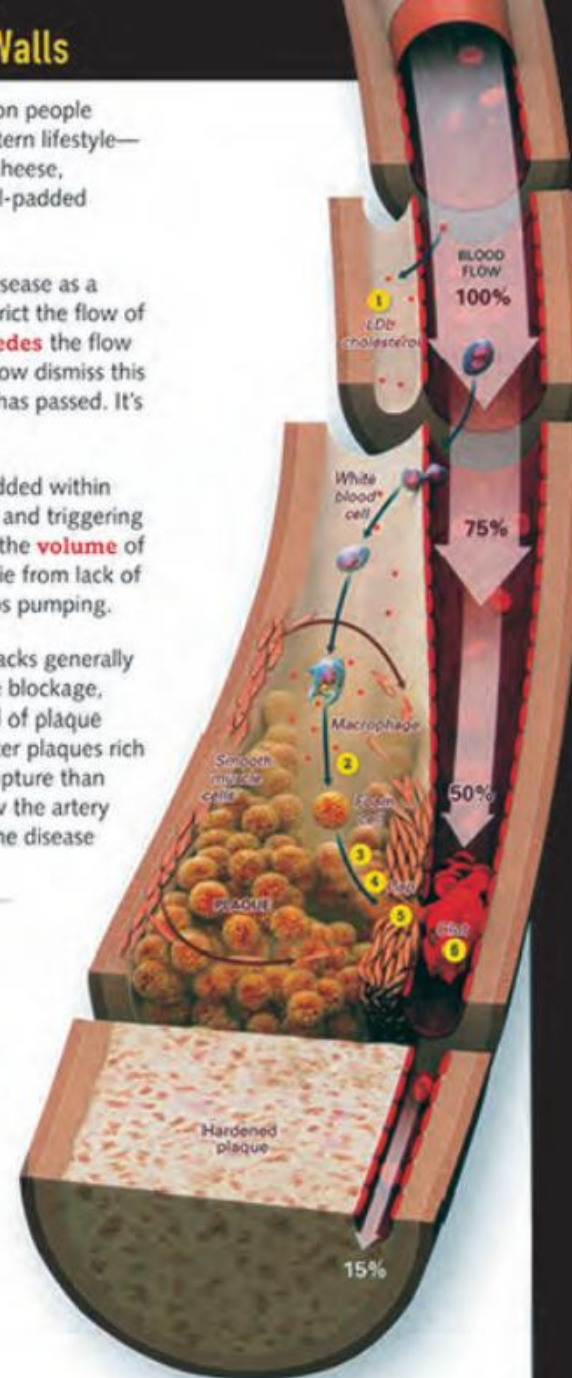
Most heart attacks are caused by plaque embedded within the artery wall that ruptures,⁹ cracking the wall and triggering the formation of a blood clot. The clot reduces the **volume** of blood flowing to the heart muscle, which can die from lack of oxygen and nutrients. Suddenly, the pump stops pumping.

Contrary to the clogged-pipes model, heart attacks generally occur in arteries that have minimal or moderate blockage, and their occurrence depends more on the kind of plaque than on the quantity. Findings suggest that softer plaques rich in cholesterol are more unstable and likely to rupture than the hard, dense plaques that extensively narrow the artery channel. But understanding the root cause of the disease will require much more research.

⁹ If something **ruptures**, it is breaks or is broken apart.

Buildup to a heart attack

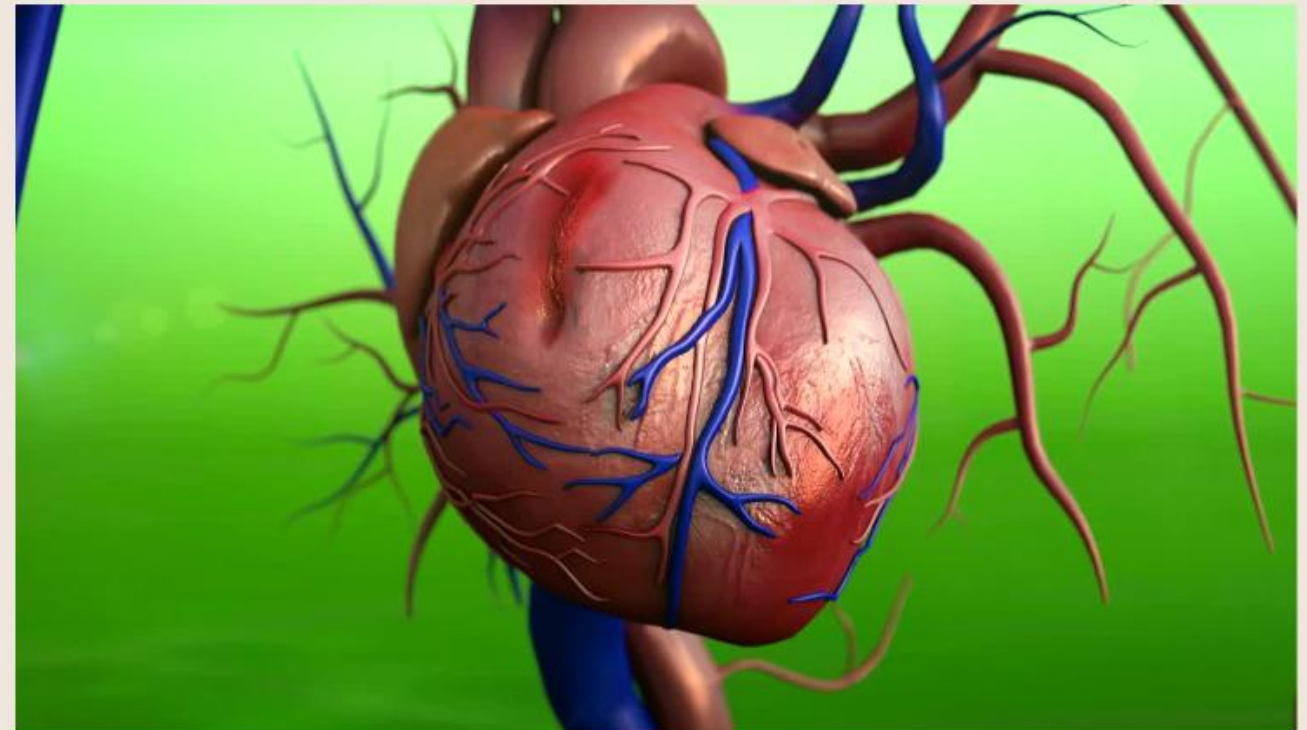
(1) Cholesterol in the blood enters the arterial wall. (2) The body’s immune system sends white blood cells (macrophages) to consume the cholesterol; (3) these grow to form foam cells, a component of plaque. (4) Muscle cells on the wall create a cap, (5) which may be weakened by the foam cells. (6) If the cap cracks, plaque moves into the bloodstream. A clot may form, restricting the volume of blood, causing a heart attack.



Read and answer

Tobacco smoking, physical inactivity, and poor diet are the main lifestyle-related factors leading to **cardiovascular** diseases (CVDs), which cause heart attacks and strokes. In the United States and Europe, CVDs have reached epidemic levels and are responsible for nearly half of all deaths. CVD-related **mortality** is also rising in the developing world, as more people become exposed to the risk factors. A diet high in cholesterol, a substance prevalent in many fatty foods, is a leading cause of **coronary** heart disease. High levels of cholesterol can cause plaque to form in **arteries**. This eventually leads to atherosclerosis (arterial blockage), which restricts the flow of blood and causes heart attacks. In addition to lifestyle factors, scientists are looking at how other factors, including genetics, may also contribute to the high incidence of CVDs.

1. What are the main factors affecting the health of the heart?
2. In your own words, explain the process that can lead to heart disease.





LET'S FIND OUT MORE!



Reading Comprehension

Multiple Choice. Choose the best answer for each question.

Detail

1. According to the information in paragraph F, most people do not realize that heart disease _____.
 - a. begins quite early in life
 - b. can be predicted in most cases
 - c. kills more people than wars or car accidents
 - d. can be prevented by eating salad and high-fiber cereal

Detail

2. What does the author say about dysfunctional *MEF2A*?
 - a. It is not common, but it is very dangerous.
 - b. It cannot be detected with current technology.
 - c. It can cause problems other than heart disease.
 - d. It is less dangerous than most genetic mutations.

Vocabulary

3. The phrase *tracking down* in line 65 is closest in meaning to _____.
 - a. taking apart
 - b. classifying
 - c. searching for
 - d. isolating

Reference

4. The word *ones* in line 75 refers to _____.
 - a. doctors
 - b. genes
 - c. statins
 - d. patients

Rhetorical Purpose

5. The purpose of the statistics in paragraph I is to _____.
 - a. warn readers not to ignore the signs of heart disease
 - b. explain how it is possible to misinterpret certain symptoms
 - c. show why risk assessment is vital for preventing heart disease
 - d. prove that heart disease is more dangerous to women than to men

Inference

6. The author implies that the drugs known as statins _____.
 - a. may cause liver damage
 - b. will help patients lose weight
 - c. are used to treat muscle pains
 - d. produce results that are easily measured

Main Idea

7. The main idea of the section on page 58 is that _____.
 - a. doctors are unsure exactly why plaque builds up inside artery walls
 - b. the "clogged-pipes model" does not fully explain why heart attacks occur
 - c. the export of Western lifestyle has caused a sudden increase in heart disease
 - d. a diet high in cholesterol is the biggest single factor contributing to heart disease

Critical Thinking

Analyzing: The author indicates that genetic factors can contribute to the possibility of heart disease. What evidence is given to show that this is true?

Discussion: Imagine that, in the future, there will be a medical test that can accurately predict your life expectancy. Would you want to know that information? Why or why not?



Multiple Choice: Choose the best answer for each question.

Detail

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Scanning. Quickly scan the article to find answers to these questions. Write each answer and note the line number(s) where you found it.

1. A single gene might contribute what percentage to a person's total risk of heart disease?

_____ line(s) _____

2. What would a comprehensive blood test for coronary risks show?

_____ line(s) _____

3. How did Don Steffensen first know about the genetics research?

_____ line(s) _____

4. What can an angiography test show?

_____ line(s) _____

5. What is Stephen Ellis collecting from his patients?

_____ line(s) _____

6. Where does Eric Topol work?

_____ line(s) _____

7. How did Don Steffensen survive his heart attack?

_____ line(s) _____

8. People with what condition would be advised to take statins?

_____ line(s) _____

Scanning for details

Play in teams and compare

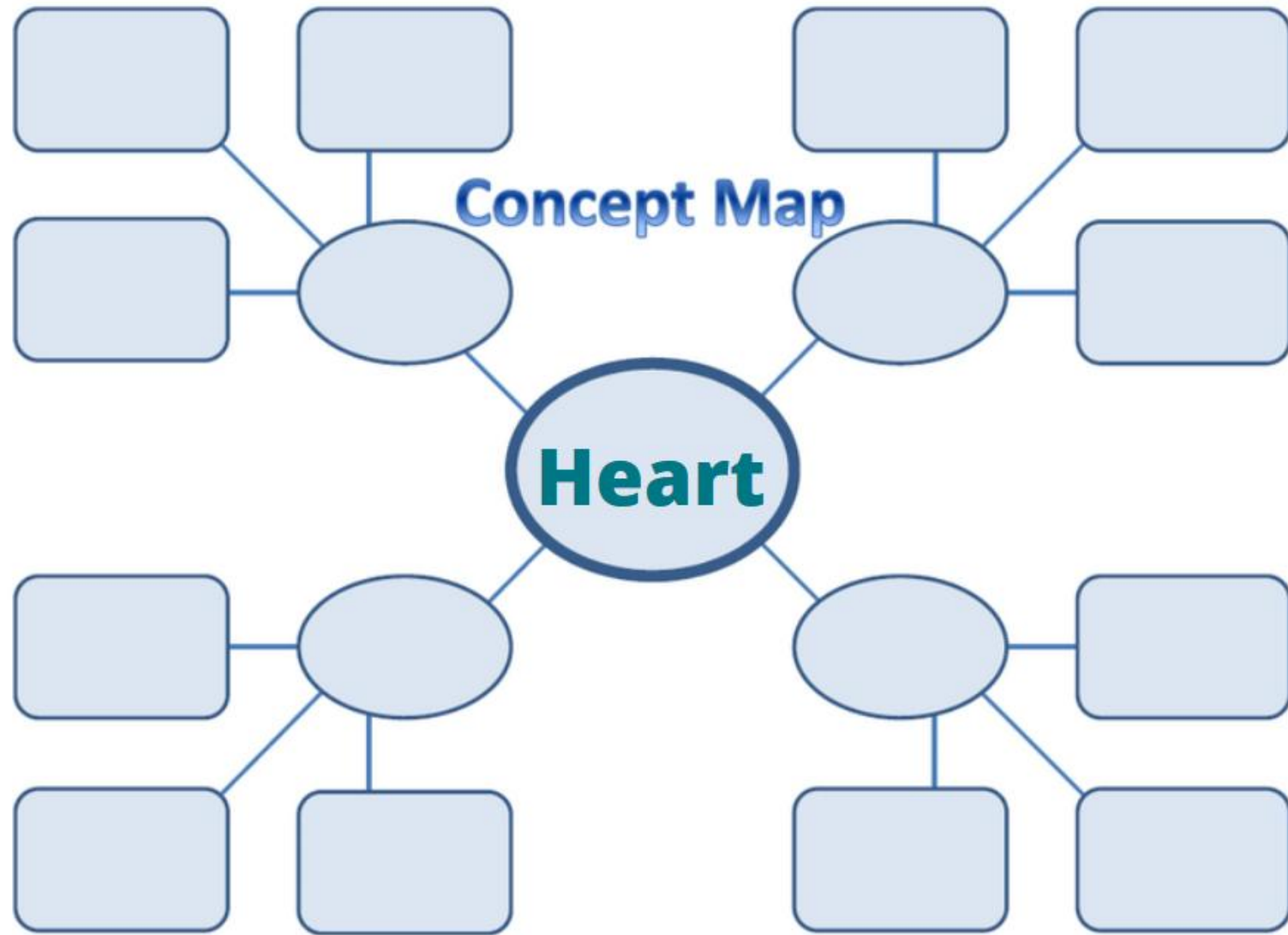
Scanning. Quickly scan the article to find answers to these questions. Write each answer and note the line number(s) where you found it.

1. A single gene might contribute what percentage to a person's total risk of heart disease?
_____ line(s) **48**
2. What would a comprehensive blood test for coronary risks show?
_____ line(s) **68-70**
3. How did Don Steffensen first know about the genetics research?
_____ line(s) **18-21**
4. What can an angiography test show?
_____ line(s) **88-92**
5. What is Stephen Ellis collecting from his patients?
_____ line(s) **54-55**
6. Where does Eric Topol work?
_____ line(s) **23-26**
7. How did Don Steffensen survive his heart attack?
_____ line(s) **9**
8. People with what condition would be advised to take statins?
_____ line(s) **96-98**

Scanning for details

Play in teams and compare

**IN GROUPS, COMPLETE THE CONCEPT MAP ACCORDING TO
THE PASSAGE**





Kahoot!

**THANK YOU FOR YOUR
ATTENTION**



YESSSS, FINALLY OVER!